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Lyapunov-guided cooperative games enable stable constraint fusion in LLM-based multi-agent autonomous systems

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In large language model-based multi-agent systems, the constraints of individual agents are mutually coupled and evolve dynamically throughout the generation process during collaborative decision-making. Without a convergence mechanism, the system state tends to oscillate among multiple local solutions, failing to converge stably toward a feasible solution that satisfies all constraints. This paper formulates the constraint fusion problem as a stability convergence problem of dynamical systems and proposes a Lyapunov-guided cooperative differential game framework. The framework integrates multi-dimensional constraint deviations into a unified system state metric via a Lyapunov function and, drawing on cooperative differential game theory, searches for approximate Pareto-improving directions within a model predictive control formulation. Real-time constraint intervention on the generation process is achieved through exponential penalty modifications to the token probability distributions of the large language model, mapping the theoretically continuous control variables to actionable discrete generation strategies. Experiments conducted on the nuScenes dataset using Llama3:8B demonstrate that the proposed framework outperforms existing baseline methods in collision rate, trajectory accuracy, and constraint satisfaction rate, with the overall constraint satisfaction rate improving by 7.0% over the best-performing baseline.

Large language model-based multi-agent systems, leveraging their semantic understanding and reasoning capabilities, have been widely applied in decision-making domains such as autonomous driving and robotic collaboration^{1,2}. Complex decision-making scenarios typically require simultaneously performing observation, reasoning, decision-making, and evaluation tasks. A single agent cannot adequately address analysis and trade-offs across all dimensions. Systems therefore decompose tasks and distribute them among multiple agents^{3,4}. In such decision-making scenarios, each agent assumes a specific functional role with its own capability boundaries. Different agents collaborate through information exchange and strategy coordination to jointly accomplish complex tasks that no single agent could handle independently. During collaboration, the output of each agent becomes the input for others. The objectives and constraints of individual agents must remain consistent throughout this interaction for the system to produce effective decisions as a whole.

In practice, the system has collective objectives at the global level, while each agent also has goals and constraints defined by its own role⁵. Collective objectives require the system output to satisfy global

performance metrics, whereas individual constraints require each agent's behavior to conform to its role specification. These two types of objectives are not always aligned. When one agent adjusts its behavior to satisfy its own constraints, this adjustment alters the input conditions for other agents and consequently affects their constraint satisfaction status. In autonomous driving, for example, when a planning agent modifies candidate trajectories to avoid obstacles, the safety evaluation agent must re-examine the collision risk of the new trajectories, and the comfort evaluation agent faces corresponding changes in acceleration and steering metrics. This influence is bidirectional: one round of adjustment triggers a new round of constraint changes, and constraint relationships continue to evolve throughout the collaboration process¹. Without an effective convergence mechanism, the system may oscillate among multiple local solutions, failing to converge stably toward a feasible solution that simultaneously satisfies all constraints^{6–8}. In safety-critical scenarios, such oscillation means that the system's output decisions become unpredictable, directly compromising system reliability and deployment feasibility^{9–12}.

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Existing large language model-based multi-agent frameworks have adopted various approaches to handle inter-agent constraint issues¹³. One common practice is to embed constraint rules directly in prompts, encoding constraints as natural language within the agent's input^{14–16}. Another approach employs hierarchical coordination structures, where upper-level agents aggregate the outputs of lower-level agents and perform conflict resolution^{17,18}. Methods such as GPT-Driver, PlanAgent, and LLM-ASSIST have made progress along these directions in autonomous driving scenarios. These methods evaluate the constraint satisfaction of each agent within each interaction round but do not track how constraints propagate across agents, nor do they model the evolution of constraint states throughout the generation process^{19–21}. In the domain of stability theory, Neural Lyapunov methods construct Lyapunov functions via neural networks to generate stability certificates for continuous control systems²². Recent work has further introduced Lyapunov control functions into inverse reinforcement learning frameworks to learn the stability structure of car-following behavior from traffic data²³. These efforts target continuous control systems with explicit dynamical equations, which differ from the discrete text generation setting of large language models²⁴. Introducing stability guidance into the constraint fusion of large language model-based multi-agent systems remains an insufficiently explored direction.

To address this gap, a mechanism is needed that can globally assess the system's constraint satisfaction state and guide the system toward the feasible region when constraints are violated. Lyapunov stability theory provides a theoretical foundation for this purpose²⁵. A scalar function is constructed to integrate constraint deviations across different dimensions into a unified system state metric. This metric is required to decrease monotonically along the system trajectory, forming a convergence mechanism. The farther the system deviates from the feasible region, the stronger the corrective force, preventing repeated oscillation among local solutions. However, a convergence mechanism alone is insufficient to coordinate the behavior of multiple agents. Independent optimization of individual objectives by each agent may lead to conflicting directions, undermining the overall convergence. Cooperative differential game theory provides a collaborative framework for this purpose²⁶. Under this framework, all agents jointly optimize a common objective function, pursuing Pareto-improving directions that improve overall system performance without degrading any individual agent's payoff^{27,28}. Unlike non-cooperative games where Nash equilibria serve as solutions^{29,30}, the cooperative framework avoids the efficiency loss resulting from independent strategic play among agents^{31,32}. Combining the convergence principle of Lyapunov theory with the collaborative framework of cooperative differential games enables stable convergence to be maintained while multiple agents jointly optimize^{33–35}.

Unlike Neural Lyapunov methods, this paper does not construct exact stability certificates but instead employs the Lyapunov function as a design tool embedded within the multi-agent optimization framework. Real-time constraint intervention is achieved through modifications to the token probability distributions of the large language model, mapping theoretically continuous control variables to actionable discrete generation strategies. Based on this idea, this paper proposes a Lyapunov-guided cooperative differential game framework for constraint fusion. The framework comprises four modules as in Fig 1. The constraint modeling module integrates multi-dimensional constraint deviations into a system-level Lyapunov function, providing a unified state assessment basis for subsequent modules. The predictive decoding module searches for approximate Pareto-improving directions over a finite horizon via model predictive control, coordinating the generation processes of individual agents. The constraint monitoring module detects constraint violations in real time based on distance metrics and corrects the system state back to the feasible region through probabilistic projection. The incentive enhancement module maintains long-term cooperative motivation among agents through a reputation system and reward mechanisms. The four modules form a closed

loop of “modeling–prediction–monitoring–enhancement,” executed online at each token generation step.

The contributions of this paper are as follows:

- A perspective is provided for the constraint fusion problem in large language model-based multi-agent systems. Constraint handling is reformulated from static rule evaluation to a stability convergence problem of system states, enabling the convergence principles of Lyapunov stability theory to be applied to discrete text generation scenarios.
- A mapping is established between continuous control theory and the discrete generation of large language models. Through exponential penalty modifications to token probability distributions, continuous control variables in the theoretical framework are transformed into real-time interventions on the generation process, providing a reusable interface design for future research.
- Experiments conducted on the nuScenes dataset using Llama3:8B demonstrate that the proposed framework outperforms existing baseline methods in collision rate, trajectory accuracy, and constraint satisfaction rate, with the overall constraint satisfaction rate improving by 7.0% over the best-performing baseline.

Results

Data source and processing

This study uses the nuScenes dataset for experimental validation. The dataset is designed for autonomous driving scenarios and contains 1000 real-world driving sequences, equipped with 6 cameras, 1 LiDAR, and 5 millimeter-wave radars, providing approximately 1.4 million 3D annotation boxes covering 23 object categories.

To enable effective conversion of 3D environmental data for large language models, this study establishes a vehicle-centric polar coordinate system that spatially divides the surrounding environment into 8 directional sectors and 3 distance levels. Through a semantic mapping mechanism, numerical annotations are converted into natural language descriptions, constructing a unified environmental state description framework that integrates multi-dimensional information, including weather conditions, road surface types, and obstacle distributions.

The complete data processing pipeline comprises five stages. First, 3D annotation information is extracted from keyframes of the nuScenes data. The global coordinate system is then transformed into a local coordinate system centered on the ego vehicle. Objects are subsequently assigned to corresponding spatial regions based on predefined angle and distance boundaries. Temporal frame information is then used to compute object motion states. Finally, structured natural language environmental descriptions are generated.

Regarding computational infrastructure, this study employs Llama3:8B as the core language model, deployed using the Ollama framework. The experimental platform is built on a workstation equipped with dual NVIDIA GeForce RTX 3090 GPUs, each with 24GB of video memory. The experiments use the complete nuScenes validation set, comprising 150 driving scenarios with approximately 6000 keyframe samples. The generation process is executed online at each token step, with constraint monitoring and projection integrated in real time into the generation pipeline. The detailed procedure is provided in Supplementary Note 3.

Evaluation metrics

This study combines standard evaluation metrics from UniAD and ST-P3 to assess system performance from three perspectives: path quality, safety, and constraint compliance. Path quality is measured by the L2 deviation metric, which computes the average Euclidean distance between corresponding points on the generated path and the ground-truth trajectory. Safety is measured by the collision rate metric, which detects the proportion of conflicts between the vehicle traveling along the planned trajectory and environmental obstacles. Constraint compliance is measured by three

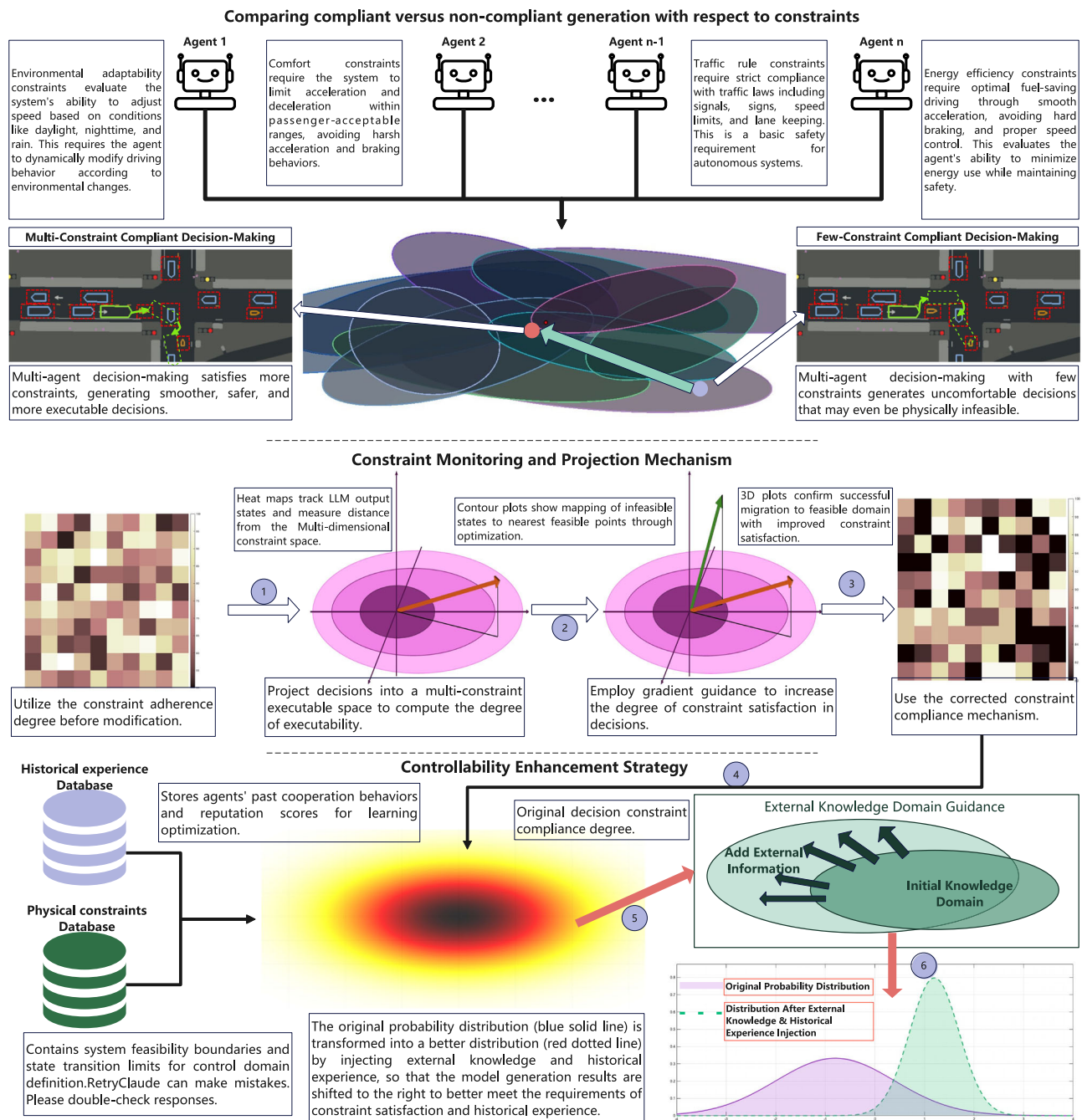


Fig. 1 | Overview of the Lyapunov stability-based multi-agent constraint fusion framework. This framework addresses multi-agent constraint fusion for autonomous driving systems. The top panel compares multi-constraint compliant decision-making (left), which generates smoother, safer, and more executable decisions, with few-constraint compliant decision-making (right), which produces uncomfortable and potentially infeasible decisions. The central 3D visualization illustrates the constraint monitoring and projection mechanism: (1) heat maps track LLM output states and measure distances from multi-dimensional constraint space, (2) contour plots show mapping optimization from infeasible to nearest feasible

points, and (3) 3D plots confirm successful migration to feasible domains through improved constraint satisfaction. The bottom section presents the controllability enhancement strategy, incorporating historical experience databases for agent cooperation behaviors, physical constraint databases for feasibility boundaries, and external knowledge domain guidance that transforms the original probability distribution (blue solid line) into a better constraint-satisfying distribution (red dotted line). The framework ensures system stability through Lyapunov functions and achieves Pareto-optimal strategies via cooperative differential games.

constraint satisfaction rates corresponding to the speed constraint, speed stability constraint, and following distance constraint, all expressed as time-averaged satisfaction rates. Detailed definitions and formulas for each metric are provided in Supplementary Note 4. The specific constraint parameters used in the experiments are: maximum speed $v_{\max} = 70$ km/h, speed variance threshold $\sigma_v^2 = 12.5$ (km/h)², and minimum following distance $d_{\min} = 5$ m.

Overall performance comparison

Table 1 and Fig. 2 present the comprehensive performance evaluation results. Experiments are compared on the ST-P3³⁶ and UniAD³⁷ evaluation platforms. Baseline methods include both traditional planning methods and large language model-driven methods. Among traditional methods, ST-P3 (Spatial-Temporal Perception-Prediction-Planning) is an end-to-end vision-based driving framework, VAD³⁸ (Vectorized Autonomous

Table 1 | Comprehensive performance evaluation results

Method	Collision rate (%)			L2 error (m)		
	1 s	2 s	3 s	1 s	2 s	3 s
ST-P3 platform						
ST-P3	0.23	0.62	1.27	1.33	2.11	2.90
VAD	0.07	0.10	0.24	0.17	0.34	0.60
GPT-driver	0	0.0789	0.3667	0.1837	0.407	0.7549
Ours-cooperative	0.0134	0.0671	0.3117	0.1612	0.3796	0.7016
UniAD platform						
NMP	-	-	1.92	-	-	2.31
SA-NMP	-	-	1.59	-	-	2.05
FF	0.06	0.17	1.07	0.55	1.20	2.54
EO	0.04	0.09	0.88	0.67	1.36	2.78
GPT-driver	0	0.275	1.2178	0.2556	0.7862	1.7048
Ours-cooperative	0.0156	0.2338	1.0351	0.2223	0.7181	1.5566

Comparison is conducted on the ST-P3 and UniAD evaluation platforms. Bold values indicate the best performance among LLM-based methods on each platform.

Driving) uses vectorized scene representations for planning, NMP³⁹ (Neural Motion Planner) performs path selection through neural network-based cost volume estimation, SA-NMP⁴⁰ (Safety-Aware NMP) adds safety constraints to NMP, FF (Freospace) plans based on free space, and EO (Ego-Only) plans using only ego vehicle information. Among large language model methods, GPT-Driver reformulates motion planning as a language modeling problem.

On the ST-P3 platform, the proposed method achieves a three-horizon average collision rate of 0.131%, compared to 0.149% for GPT-Driver, representing a reduction of 11.9%. The average L2 error is 0.414 m, compared to 0.449 m for GPT-Driver, a reduction of 7.1%. On the UniAD platform, the average collision rate is 0.428%, compared to 0.498% for GPT-Driver, a reduction of 13.9%. The average L2 error is 0.832 m, compared to 0.916 m for GPT-Driver, a reduction of 9.1%. As the prediction horizon extends from 1 s to 3 s, the collision rate growth of the proposed method is comparable to that of GPT-Driver, with no significant performance degradation. The above results are averages over multiple independent runs, and the differences between the proposed method and each baseline are statistically significant at the $p < 0.05$ level.

Constraint satisfaction rate comparison

Table 2 presents the constraint satisfaction rate evaluation results. The comparison methods include four large language model-driven baselines, all of which handle constraints by embedding constraint rules in prompts. GPT-Driver⁴¹ implements motion planning through language modeling, PlanAgent⁴² employs a hierarchical planning agent architecture, LimSim⁴³ integrates LLM decision-making within a simulation environment, and LLM-ASSIST⁴⁴ uses LLMs to assist traditional planners.

The proposed method achieves a speed constraint satisfaction rate of 0.789, an improvement of 6.9% over LLM-ASSIST. The speed stability satisfaction rate is 0.813, an improvement of 7.1% over LLM-ASSIST. The following distance satisfaction rate is 0.687, an improvement of 7.2% over LLM-ASSIST. The overall constraint satisfaction rate is 0.763, an improvement of 7.0% over LLM-ASSIST. Among the three constraints, the following distance satisfaction rate is the lowest, reflecting the difficulty of maintaining safe distances in dynamic traffic environments.

Ablation study

To analyze the contribution of each module in the framework, this study designs the following ablation variants. Projection-only retains only the

constraint projection module without the cooperative game structure or the Lyapunov penalty. Game-only retains the cooperative game structure for joint optimization but removes the Lyapunov stability term. Full denotes the complete framework. Table 3 reports the ablation results, and Fig. 3 shows the corresponding Lyapunov function trajectories.

The Projection-only variant achieves an overall constraint satisfaction rate of 0.712, which is 5.1 percentage points lower than the full framework. This variant independently corrects each agent's constraint violation at each round but lacks inter-agent coordination, resulting in higher collision rates and L2 errors than the full framework. The Game-only variant achieves an overall constraint satisfaction rate of 0.731, which is 3.2 percentage points lower than the full framework. This variant coordinates agent behavior through the cooperative game structure and outperforms Projection-only in collision rate and L2 error, but the removal of the Lyapunov stability term leads to a decrease in constraint satisfaction rate. The full framework achieves the best results across all three metrics, indicating that both the cooperative game structure and Lyapunov guidance contribute to system performance, and their combination is superior to either used alone.

Stability analysis

To directly validate the convergence effect of the Lyapunov guidance mechanism, this study records the trajectory of the system-level Lyapunov function $V(x)$ over generation steps during the generation process. Fig. 3 shows the $V(x)$ curves of the three ablation variants under the same scenario.

In the representative scenario, the $V(x)$ curve of the Full framework exhibits a near-monotonic decreasing trend, converging to a low level within approximately 15 generation steps. The $V(x)$ curve of the Game-only variant shows an overall decreasing trend but with local fluctuations, and its convergence speed is slower than that of the Full framework. The $V(x)$ curve of the Projection-only variant exhibits pronounced oscillations, with repeated increases and decreases over multiple time intervals, reflecting the system state jumping among local solutions in the absence of global coordination. The final steady-state $V(x)$ values of the three variants are 0.038 for Full, 0.057 for Game-only, and 0.089 for Projection-only; the Full framework converges to the lowest level. These results indicate that the Lyapunov stability term provides an effective convergence mechanism for the system, and the cooperative game structure further improves the smoothness of convergence.

Discussion

The proposed framework addresses different aspects of constraint fusion through its four modules. The constraints of different agents lack a unified metric in the semantic space; the Lyapunov function integrates these heterogeneous constraints into a single scalar value, enabling the system to assess the degree of deviation of the current state with a single quantity. The constraints among agents are mutually coupled, as one agent's adjustment alters the constraint boundaries of others; the model predictive control framework identifies this propagation effect through finite-horizon lookahead and intervenes before conflicts arise. The Lyapunov stability condition requires this scalar value to decrease monotonically along trajectories, with stronger corrections for larger deviations, providing the system with a convergence mechanism. The cooperative game framework enables all agents to jointly optimize along Pareto-improving directions, avoiding the directional conflicts that arise from independent adjustments. Shapley values and the reputation mechanism sustain cooperative motivation among agents over long-term interactions.

The ablation study results are consistent with the above analysis. The Projection-only variant independently corrects each agent's deviation at each round, achieving an overall constraint satisfaction rate of 0.712. This variant can rectify individual constraint violations, but when one agent's correction triggers new deviations in other agents, the system lacks coordination mechanisms. With the addition of the cooperative game structure, the Game-only variant improves the constraint satisfaction rate to 0.731, as joint optimization among agents reduces mutual interference during the correction process. The full

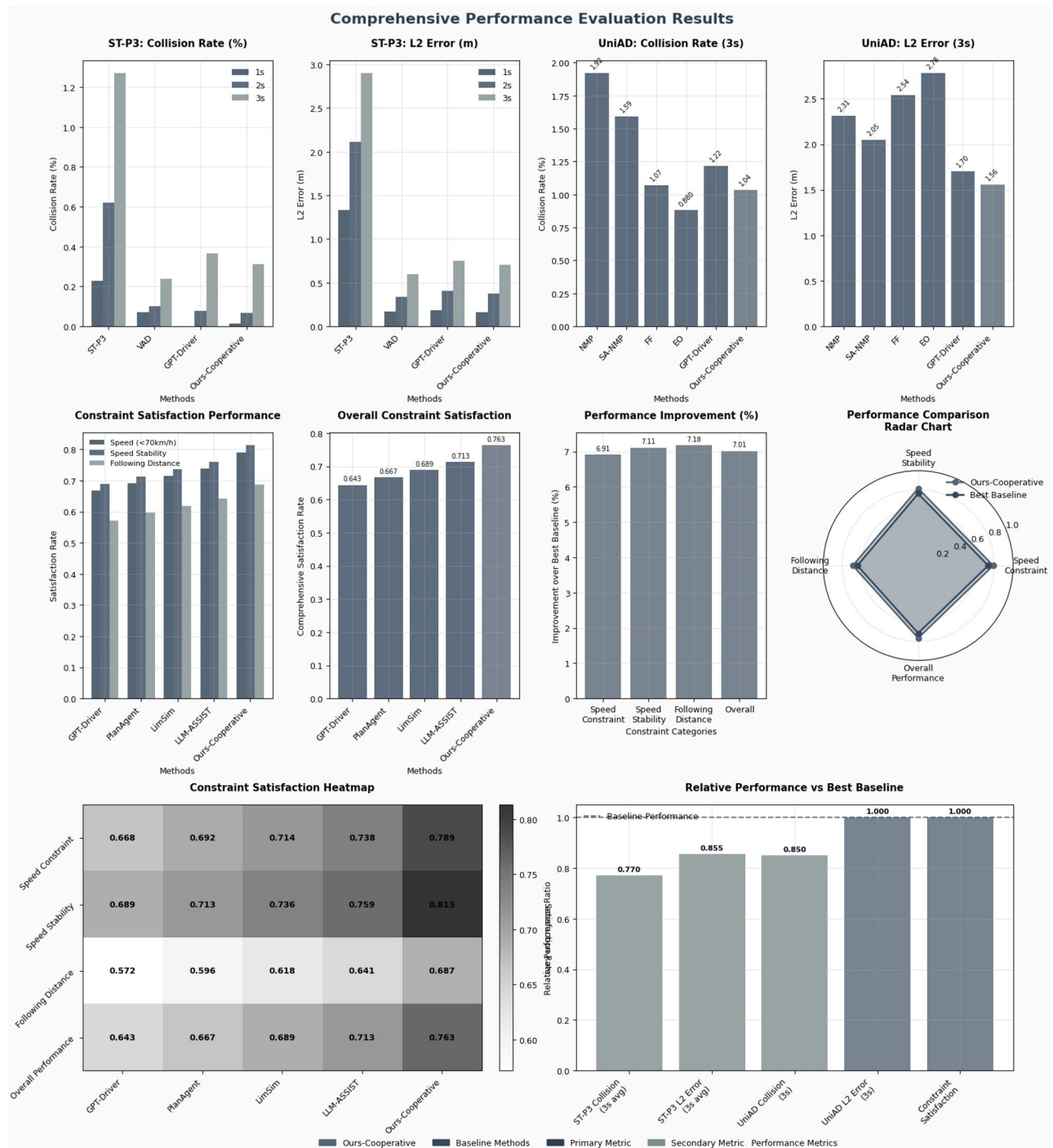


Fig. 2 | Comprehensive performance evaluation results. *Top row:* Collision rate and L2 error comparisons on the ST-P3 and UniAD platforms across 1 s, 2 s and 3 s prediction horizons. *Middle row (left to right):* Per-constraint satisfaction rates across methods; overall constraint satisfaction rates; percentage improvement of the proposed method over the best baseline (LLM-ASSIST) for each constraint category; radar chart comparing Ours-Cooperative against the best baseline across four

evaluation dimensions. *Bottom row:* Constraint satisfaction heatmap showing all methods across all constraint categories, where darker shading indicates higher satisfaction; relative performance ratio of Ours-Cooperative to the best baseline on each metric, where values at or above 1.0 indicate that the proposed method matches or exceeds the best baseline.

framework further adds the Lyapunov stability term on this basis, raising the constraint satisfaction rate to 0.763. The $V(x)$ curves in Fig. 3 illustrate the differences among the three variants. The full framework's curve exhibits a near-monotonic decrease, while the Projection-only variant's curve shows repeated increases and decreases over multiple time intervals. This contrast demonstrates that Lyapunov guidance

provides the system with the ability to recover from oscillation among local solutions, and the cooperative game structure improves the smoothness of the convergence process.

The present work may inform future research from several perspectives. For researchers working on large language model-based multi-agent systems, this paper reformulates constraint handling from

Table 2 | Constraint satisfaction rate evaluation results

Method	Speed (<70 km/h)	Stability (Var < 12.5)	Following dist. (>5 m)	Overall
GPT-driver	0.668	0.689	0.572	0.643
PlanAgent	0.692	0.713	0.596	0.667
LimSim	0.714	0.736	0.618	0.689
LLM-ASSIST	0.738	0.759	0.641	0.713
Ours- cooperative	0.789	0.813	0.687	0.763

All comparison methods handle constraints by embedding rules in prompts. Bold values indicate the best performance.

Table 3 | Ablation study results

Variant	Collision rate 3 s (%)	L2 error 3 s (m)	Overall CSR
Projection-only	0.385	0.748	0.712
Game-only	0.352	0.729	0.731
Full	0.312	0.702	0.763

Projection-only uses constraint projection alone; Game-only adds cooperative game structure; Full is the complete framework.

per-round independent evaluation to a dynamic convergence problem across generation steps. This perspective enables stability tools from control theory to be applied to discrete text generation scenarios. Future work may explore the application of other control-theoretic tools in LLM systems along this direction. For researchers in control theory, the mapping established in this paper between continuous control variables and token probability distributions represents an attempt to extend classical methods to discrete generation systems. The design of this mapping does not depend on a specific LLM architecture and can be adapted to different models. For engineering practice oriented toward real-world deployment, the four modules in the framework adopt a composable design. The constraint projection module can be independently integrated into existing systems as a safety layer, while the cooperative game structure and incentive mechanisms can be selectively introduced depending on the complexity of the scenario.

This study has the following limitations. First, the Lyapunov function is employed as a design tool in this paper, without providing formal stability theorems or convergence rate proofs. Establishing more rigorous theoretical guarantees under specific assumptions is one direction for future work. Second, the experiments validate the framework with four agents in autonomous driving scenarios, and its performance in other decision-making domains and larger-scale agent systems remains to be investigated. Third, the continuous dynamics is an approximation of the token-by-token generation process of LLMs, and the degree to which this approximation holds may vary across different types of generation tasks.

Method

Problem definition

This section formalizes the multi-agent constraint fusion problem by defining the system state, constraint structure, and optimization objective, and clarifying the concrete semantics of theoretical symbols in the LLM generation setting, as in Fig. 4.

Consider a collaborative system comprising N agents $\mathcal{A} = \{A_1, A_2, \dots, A_N\}$. The prompt of each agent A_i consists of three components—role description, style guidance, and constraint rules—which jointly define its behavioral pattern and responsibility boundaries. The state

of agent A_i at time t is defined as

$$s_i^t = (\text{TokenSeq}_{i,t}, \text{Context}_{i,t}, \text{Prompt}_i, \text{MsgHist}_{i,t}) \quad (1)$$

In the LLM-based implementation, s_i^t determines the conditional token probability distribution $P(y_{i,t}|\text{context})$ of agent A_i at time t and its generation behavior. TokenSeq corresponds to the sequence of tokens already generated, Context corresponds to the current attention state, and MsgHist corresponds to the textual messages exchanged among agents.

Each agent must simultaneously satisfy individual constraints and system-level constraints:

$$g_i(s_i^t, \text{Prompt}_i) \leq 0 \text{ and } h_{\text{global}}(\mathbf{S}_t) \leq 0 \quad (2)$$

where the global system state $\mathbf{S}_t = [s_1^t, \dots, s_N^t]^T$ is the composition of all agent states. The individual constraint g_i characterizes the degree to which an agent adheres to its own role specification, encompassing multiple dimensions including role consistency, style consistency, and communication coordination, denoted as $\{g_{i,c}\}_{c=1}^{C_i}$. The system constraint h_{global} characterizes the coordination consistency requirements among agents. These two types of constraints are in tension: individual constraints pursue the distinctiveness and specialization of each role, while system constraints demand coordinated unity.

The constraint coupling manifests at three levels. Role-level constraint conflicts arise from the differentiated requirements of different agent roles. Coordination-level constraint conflicts arise from the complexity of inter-agent communication and collaboration. System-level constraint conflicts arise from the contradiction between individual rationality and collective rationality. When agent A_i adjusts its behavior to satisfy its individual constraints, this affects the constraint satisfaction of other agents, which in turn affects the global constraints of the system.

To uniformly characterize the interaction effects among these multi-level constraints, the constraint fusion function is defined as

$$\psi_{\text{multi}}(\mathbf{S}_t) = f(g_1(s_1^t), g_2(s_2^t), \dots, g_N(s_N^t), h_{\text{global}}(\mathbf{S}_t), \text{InteractionEffects}) \quad (3)$$

Let the generation sequence be $\mathbf{Y} = \{\mathbf{y}_1, \dots, \mathbf{y}_T\}$, where $\mathbf{y}_t = [y_{1,t}, \dots, y_{N,t}]^T$ denotes the joint output of all agents at time t . The optimization objective of multi-agent constraint fusion is defined as

$$\max_{\mathbf{Y}} \sum_{t=1}^T \sum_{i=1}^N \log P(y_{i,t} | \text{TokenSeq}_{i,<t}, \text{Prompt}_i) - \lambda \psi_{\text{multi}}(\mathbf{S}_t) \quad (4)$$

The feasible region of the multi-agent system is defined as the state space in which all agents simultaneously satisfy constraints across all dimensions:

$$\mathcal{F} = \{\mathbf{S}_t \in \mathcal{S}^N | \forall i, c: g_{i,c}(s_i^t) \leq 0, h_{\text{global}}(\mathbf{S}_t) \leq 0\} \quad (5)$$

In the autonomous driving scenario, $\mathcal{F}$ is instantiated as the intersection of specific physical constraints, including the speed constraint $v(t) \leq v_{\text{max}}$, the speed stability constraint $\text{Var}(v) \leq \sigma_v^2$, and the following distance constraint $d(t) \geq d_{\text{min}}$. These scalar constraints constitute the operationalized definition of the feasible region used in the experiments.

During the token-level generation process, each decision step of an agent alters the system state and the constraint boundaries of other agents, causing the constraint relationships to evolve dynamically throughout the generation process. The formal definition of state transitions is provided in Supplementary Note 3. The objective of this paper is to design a framework that enables the system to maintain or converge to the feasible region $\mathcal{F}$ while optimizing generation quality.

Ablation Study and Stability Analysis

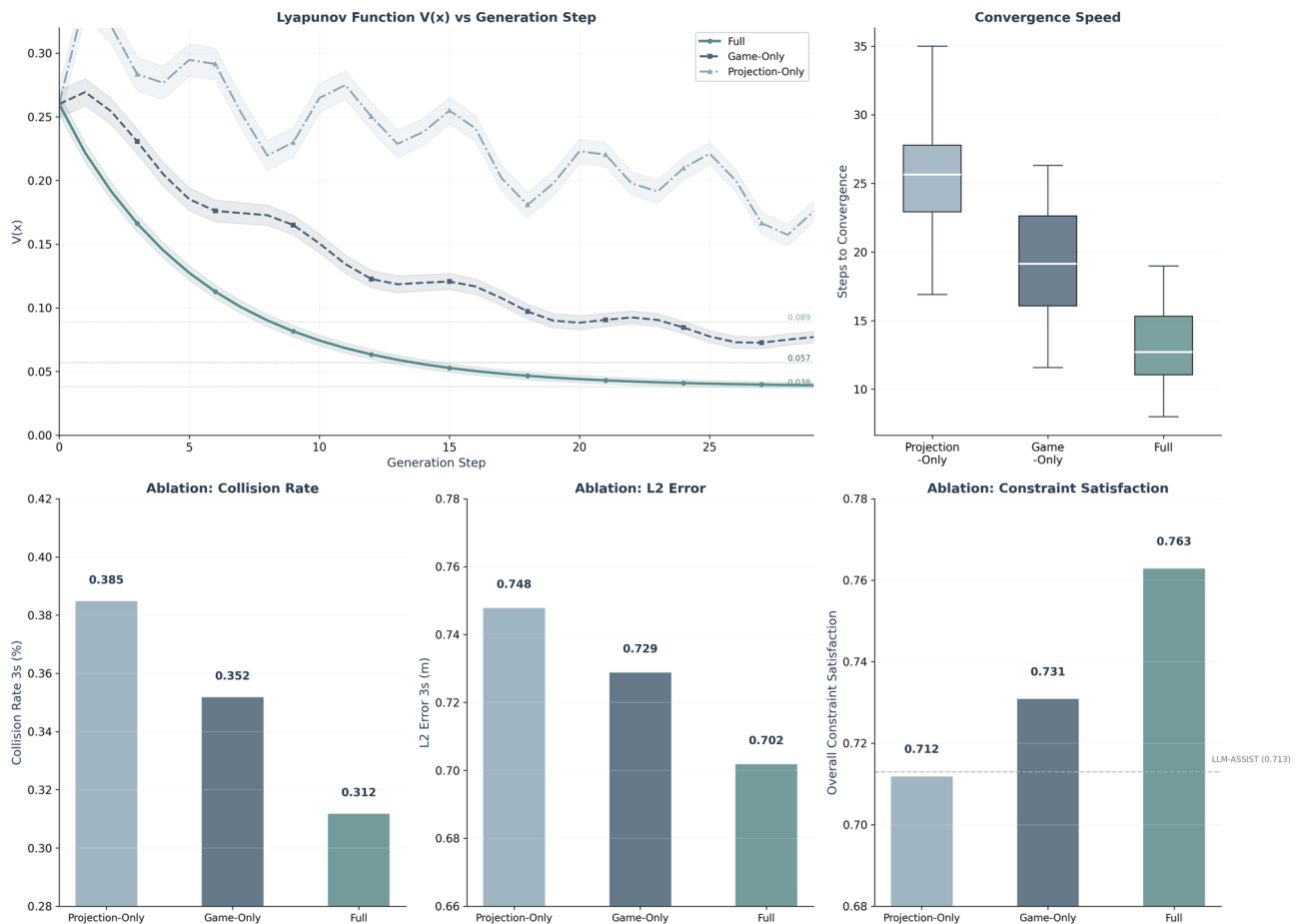


Fig. 3 | Multi-agent constraint fusion problem illustration in dynamic decision-making. This figure illustrates the core challenge of multi-agent constraint fusion in collaborative systems. The upper section depicts the autonomous driving scenario where agents must navigate complex constraint interactions: (4) mapping constraint-violating states to nearest feasible equivalents for multi-dimensional constraint satisfaction, (5) constraint monitoring that detects deviations and generates error signals for correction, and (6) closed-loop control adjusting output probabilities to favor constraint-satisfying decisions. The lower section shows the

system's temporal evolution from initial state S_1 through sequential states $S_2, S_3, \dots, S_{n-1}, S_n$, where: (1) each system state is determined by previous history and current decisions with evolution governed by specific rules, (2) only multi-dimensionally constraint-satisfying decisions are feasible and maintain future viability, and (3) forward-looking predictions decode the most feasible path for constraint satisfaction. The LLM agent integrates external knowledge bases through constraint monitoring and feasibility projection to achieve stable multi-agent coordination under complex constraint networks.

Multi-dimensional constraint feasible region modeling

Building on the problem definition above, this section models multi-agent constraint fusion as a continuous optimization problem inspired by cooperative differential games and constructs a Lyapunov function as a unified measure of system stability, as in Fig. 5.

Following the agent set $\mathcal{A}$ and state definition s_i^t from the problem definition, the system constructs a directed communication graph based on role prompt definitions and collaboration requirements, characterizing the information interaction structure and cooperation intensity among agents. This communication topology is dynamically adjusted during system operation, reflecting the complementarity of role relationships. The formal definition of the communication graph is provided in Supplementary Note 3.

To incorporate the discrete token-level generation process into a continuous optimization framework, this paper maps the agent state s_i^t from the problem definition to a vector $x_i(t)$ in a continuous state space, encompassing semantic state, context embedding, role vector, interaction history, and constraint status. The system state evolution is described by a

differential equation:

$$\frac{dx_i}{dt} = f_i(x_i(t), u_i(t), x_{-i}(t), t) \quad (6)$$

where $u_i(t)$ is the control input of agent A_i and $x_{-i}(t)$ denotes the states of other agents, capturing the dynamic coupling among agents. Here, $x_i(t)$ represents the state vector characterizing the agent's current output distribution, and $u_i(t)$ is the modulation signal applied to that distribution, corresponding in implementation to the reweighting operation on token probabilities. $x_{-i}(t)$ reflects the influence of other agents' states on the constraint boundaries of A_i . This continuous dynamics is a continuous approximation of the token-by-token generation process of LLMs. This continuous dynamics is a continuous approximation of the token-by-token generation process of LLMs.

In implementation, the continuous dynamics is discretized at the token-generation time scale. Each generation step $t \rightarrow t + 1$ corresponds to a

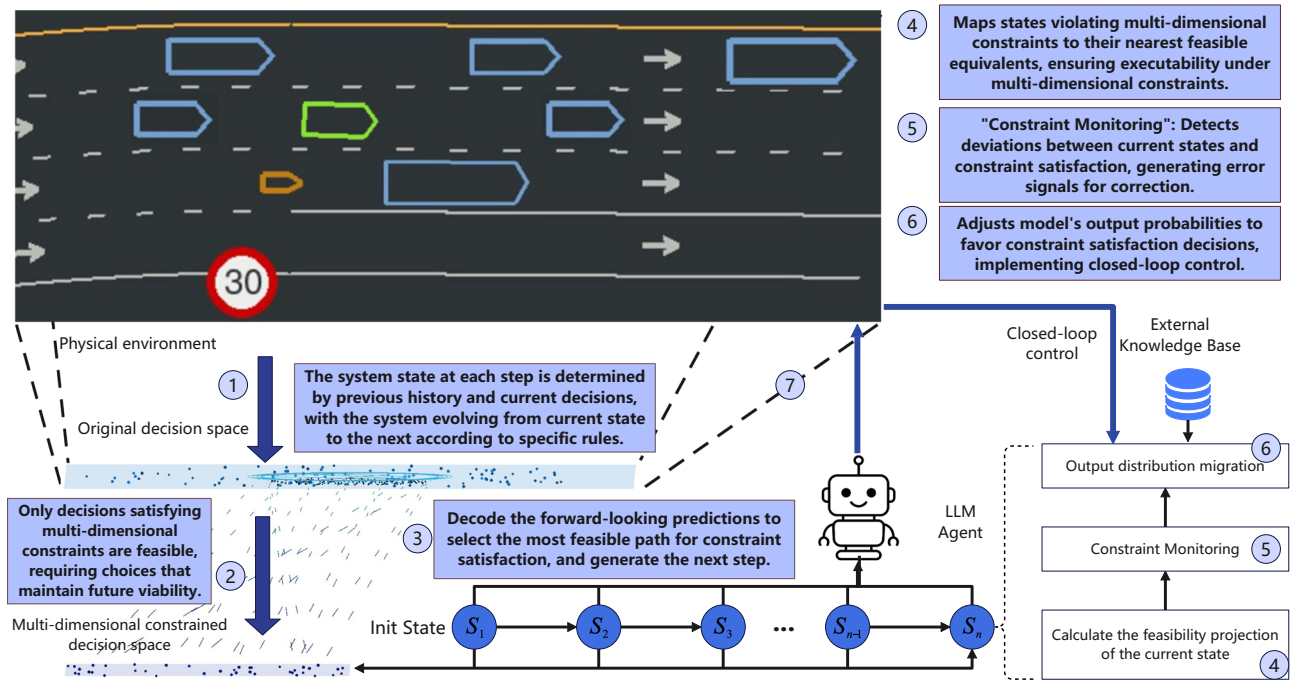


Fig. 4 | Multi-dimensional constraint feasible domain modeling for multi-agent systems. This figure illustrates the construction of multi-dimensional constraint feasible regions in cooperative differential game frameworks for autonomous driving systems. Four agents with distinct roles contribute different constraint dimensions: weather-based speed constraints (Agent 1: foggy conditions limiting maximum speed to 30 km/h), traffic regulation compliance (Agent 2: adherence to traffic laws), comfort optimization (Agent 3: acceleration limits below 3 m/s²), and energy efficiency (Agent 4: speed and acceleration within green zones). The central

3D visualization shows the intersection of these constraint domains, where colored ellipsoidal regions represent individual agent constraint spaces and the complex vector field depicts the feasible decision trajectories. The overlapping regions demonstrate how the collaborative feasible domain $\mathcal{F}$ emerges from the intersection of all agent constraints, forming the admissible state-control space where cooperative Pareto-optimal solutions exist while satisfying both individual rationality and system-wide coordination requirements.

discrete update

$$x_i^{t+1} = x_i^t + f_i(x_i^t, u_i^t, x_{-i}^t, t) \cdot \Delta t \quad (7)$$

where $\Delta t = 1$ represents one token generation step. Accordingly, the Lyapunov stability condition $dV/dt \leq -\epsilon \|x - \mathcal{F}\|^2$ is evaluated in its discrete-time form

$$V(\mathbf{x}^{t+1}) - V(\mathbf{x}^t) \leq -\epsilon \|\mathbf{x}^t - \mathcal{F}\|^2 \quad (8)$$

at each generation step. This is the form used in the stability analysis experiments, where the $V(\mathbf{x})$ trajectories reported in Fig. 6 are recorded at per-token resolution. All constraint monitoring and probabilistic projection operations described in subsequent sections are likewise executed at this discrete time scale, ensuring consistency between the theoretical formulation and the experimental evaluation.

The system-level joint objective function is designed as a weighted combination of all agents' objectives:

$$J(u_1, \dots, u_N) = \sum_{i=1}^N w_i \int_0^T L_i(x_i, u_i, x_{-i}, t) dt \quad (9)$$

where $w_i > 0$ is the cooperation weight of agent A_i and L_i is the instantaneous utility function. The weight allocation is dynamically adjusted based on fairness principles and contribution assessment.

This paper draws on the theoretical framework of cooperative differential games to model the above joint optimization as a cooperative game problem. The formal definition of the game is provided in Supplementary Note 1. In this framework, all agents collaboratively optimize the system-level objective function J , pursuing Pareto-improving directions through information sharing and strategy coordination. Unlike non-cooperative

games, where each agent optimizes independently with Nash equilibrium as the solution concept, in the cooperative framework, all agents jointly optimize a common objective function, with Pareto optimality as the solution concept—meaning no alternative strategy exists that simultaneously benefits all agents. The cooperative framework is chosen because multi-agent constraint fusion is a coordination problem rather than an adversarial one, and the efficiency loss of Nash equilibria is detrimental to overall system performance.

The instantaneous utility function L_i jointly considers generation quality, role consistency, coordination benefits, and constraint satisfaction:

$$L_i = \alpha_i Q_i + \beta_i R_i + \gamma_i C_i - \delta_i P_i \quad (10)$$

where Q_i measures the output quality of the agent, R_i evaluates the degree to which the agent's state conforms to its prompted role, C_i quantifies the coordination effectiveness among agents, and P_i penalizes behavior that deviates from constraints.

Each agent must satisfy multi-level constraints defined by its prompt, encompassing four dimensions: role consistency, style consistency, communication coordination, and system-level collaboration. Based on the above constraint definitions, the cooperative feasible region of the multi-agent system is constructed as:

$$\mathcal{F} = \{(x, u) | g_{i,c}(x, u) \leq 0, \forall i, c\} \quad (11)$$

This definition is consistent with the feasible region in the problem definition, expressed in the continuous state space.

To assess the degree to which the system state deviates from the feasible region, this paper designs a system-level Lyapunov function that integrates individual constraint deviations, inter-agent conflicts, and Pareto deviation

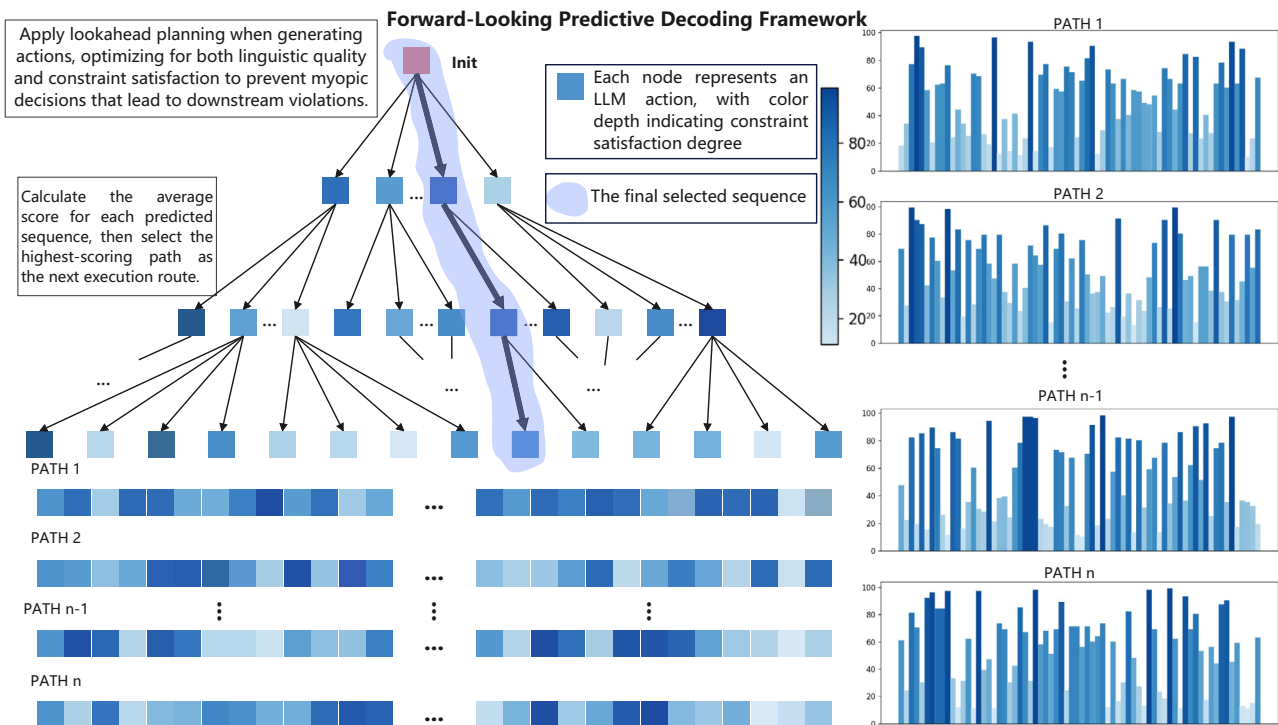


Fig. 5 | Forward-looking predictive decoding framework for multi-agent cooperative systems. This figure illustrates the cooperative differential game-driven collaborative predictive decoding mechanism. The framework employs lookahead planning to optimize linguistic quality and constraint satisfaction while preventing downstream violations. The central tree structure shows the multi-path exploration process, where each node represents an LLM action with color depth indicating confidence levels. Multiple prediction paths (PATH 1, PATH 2,..., PATH $n-1$, PATH n) are evaluated in parallel, with the system calculating average scores for each

predicted sequence and selecting the highest-scoring path as the next execution direction. The right panel displays probability distributions for different paths, demonstrating how the framework assesses and compares alternative trajectories. Through this three-stage cooperative process of collaborative optimization, information sharing, and stabilization, the system predicts future state evolution within the prediction horizon T_{pred} , solves constraint Pareto-optimal solutions, and ensures both individual role consistency and system-wide coordination through receding horizon control strategies.

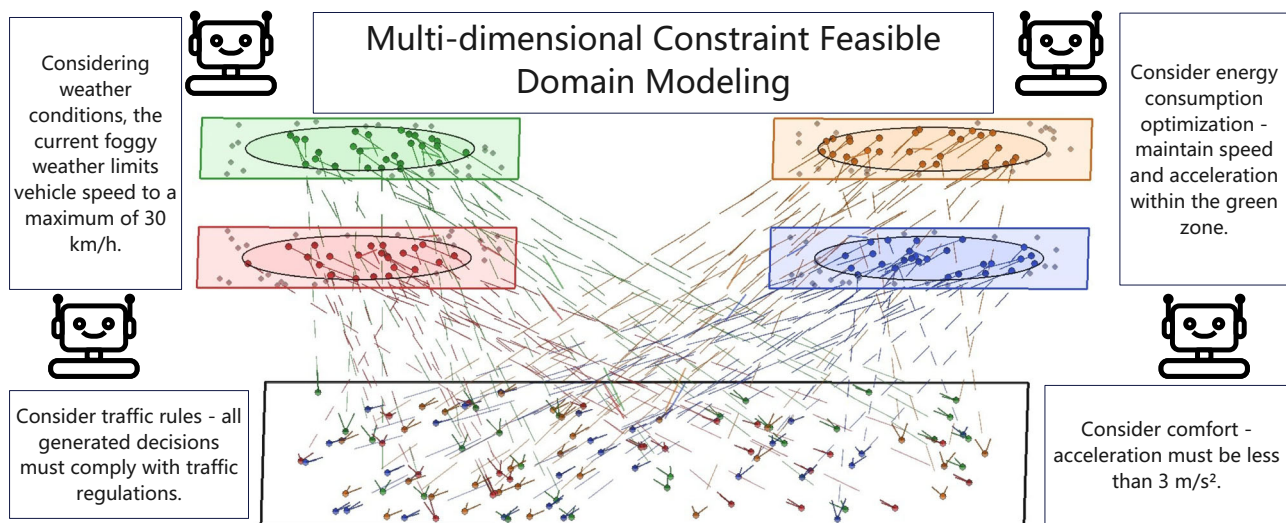


Fig. 6 | Lyapunov function $V(x)$ trajectories over generation steps for the three ablation variants. The Full framework exhibits a near-monotonic decrease, Game-only shows an overall decreasing trend with local fluctuations, and Projection-only displays pronounced oscillations.

into a single scalar measure:

$$V(x) = \sum_{i=1}^N \lambda_i V_i(x_i) + \sum_{i < j} \omega_{ij} \text{Conflict}_{ij} + \mu \cdot \text{Deviation}(x) \quad (12)$$

where $V_i(x_i)$ is the individual constraint deviation measure of agent A_i , whose specific construction is provided in Supplementary Note 1. Conflict_{ij} quantifies the degree of constraint conflict between agents i and j . $\text{Deviation}(x)$ measures the distance between the current strategy combination and the Pareto front, and $\mu > 0$ is the corresponding weight.

The design objective of the framework is to find a strategy combination $\mathbf{u}^*$ that satisfies the following Lyapunov stability condition:

$$\frac{dV}{dt} \Big|_{\mathbf{u}=\mathbf{u}^*} \leq -\epsilon \|\mathbf{x} - \mathcal{F}\|^2 \quad (13)$$

This condition requires that under the selected strategy, the Lyapunov function decreases monotonically along the system trajectory, with the rate of decrease proportional to the distance of the system from the feasible region, where $\|\mathbf{x} - \mathcal{F}\|$ denotes the distance from the system state $\mathbf{x}$ to the feasible region $\mathcal{F}$. The greater the deviation, the stronger the corrective force, thereby guiding the system state to converge toward the feasible region.

Cooperative game-driven predictive decoding

The previous section established the stability measure of the system. This section translates it into an actionable decoding strategy. Within a finite prediction horizon, an approximate Pareto-improving direction is searched through a model predictive control framework inspired by cooperative differential games, coordinating the generation processes of individual agents as in Fig. 6.

At step t , the system defines a prediction horizon length T_{pred} based on the current global state $\mathbf{x}(t)$ and the multi-agent cooperative feasible region $\mathcal{F}$, and predicts the future control trajectory $\mathbf{u}_{[t, t+T_{\text{pred}}]}$. The state evolution within the prediction horizon follows the same continuous dynamics equation as in the previous section.

The decoding process at each time step comprises three stages. In the first stage, the system determines the cooperation weights $\{w_i^t\}$ and responsibility allocation for each agent based on the current state and constraint conditions. In the second stage, all agents jointly optimize control inputs over the prediction horizon based on shared information. In the third stage, the strategy is verified and corrected through the Lyapunov condition to drive the system state toward the feasible region.

The joint objective function comprehensively considers overall system generation quality, role constraint satisfaction, collaboration benefits, and stability, and is defined as:

$$J(\mathbf{u}) = \sum_{i=1}^N w_i^t \int_t^{t+T_{\text{pred}}} L_i(\tau) d\tau + \int_t^{t+T_{\text{pred}}} \Phi(\tau) d\tau \quad (14)$$

where $L_i(\tau)$ is the instantaneous utility function of agent i , characterizing the local generation quality and constraint satisfaction of an individual agent. $\Phi(\tau)$ is the system-level cross-agent coordination term, characterizing inter-agent collaboration benefits and global consistency, complementary to L_i . L_i focuses on individual performance, while Φ focuses on overall coordination. The weight parameters are dynamically adjusted according to negotiation outcomes and historical performance.

The system searches for approximate Pareto-improving directions that satisfy the feasible region constraints within the prediction horizon:

$$\mathbf{u}^* = \arg \max_{\mathbf{u}} J(\mathbf{u}) \text{ s.t. } (\mathbf{x}(\tau), \mathbf{u}(\tau)) \in \mathcal{F}, \forall \tau \quad (15)$$

The theoretical optimality conditions of this optimization problem can be characterized by the Hamilton–Jacobi–Bellman equation; the detailed derivation is provided in Supplementary Note 1. In practice, the system employs a model predictive control framework for an approximate solution rather than computing the complete Pareto set exactly.

When multiple candidate solutions exist, the system uses the Lyapunov function value and a fairness metric as selection criteria:

$$\mathbf{u}^{**} = \arg \min_{\mathbf{u}^* \in \text{ParetoSet}} [V(\mathbf{x}) + \lambda \cdot \text{FairnessViolation}(\mathbf{u}^*)] \quad (16)$$

That is, among candidate strategies, the system preferentially selects the one that yields a lower Lyapunov function value and better fairness.

To support distributed implementation, the system decomposes the joint optimization problem into local optimization subproblems for each agent. The local optimization problem of agent i is:

$$u_i^* = \arg \max_{u_i} [w_i L_i + \sum_{j \neq i} \rho_{ij} L_j] \quad (17)$$

where ρ_{ij} is the attention weight of agent i toward the objective of agent j . Each agent optimizes its own objective while accounting for the objectives of other agents weighted by ρ_{ij} , ensuring that local decisions remain aligned with the global optimization direction.

The optimization results are realized by adjusting the generation probability distribution of each agent. For agent i , the modified token probability distribution is:

$$P_i(u_i) = \frac{P(u_i) \cdot \exp(-\beta_i \Phi_i(u_i))}{\mathcal{Z}_i} \quad (18)$$

where Φ_i is a composite measure integrating role constraints, style constraints, and collaboration benefits. This probability modification operation is the concrete implementation of the control input $u_i(t)$ in the theoretical framework. By applying an exponential penalty based on the constraint violation measure Φ_i to the original token probability distribution $P(u_i)$ of the LLM, real-time intervention on the generation process is achieved. β_i controls the intervention intensity, and $\mathcal{Z}_i$ is the normalization constant. The system repeats this process in a receding horizon fashion, updating the state and re-solving at each time step, with weights dynamically adjusted based on historical performance.

Constraint monitoring and projection

Predictive decoding provides forward-looking constraint optimization, but the prediction horizon is finite, and the constraint environment is complex, so the system still requires a real-time monitoring and correction mechanism. This section designs a constraint monitoring and probabilistic projection method based on distance metrics, serving as a complement to predictive decoding by correcting the system state back into $\mathcal{F}$ when it deviates from the feasible region.

Given the current global state $\mathbf{x}(t)$, if $\mathbf{x}(t) \notin \mathcal{F}$, the system adjusts each agent's control input $\mathbf{u}(t)$ through distance metrics and projection operations, driving the system state $\mathbf{x}(t + \Delta t)$ back to the feasible region.

The system defines a constraint distance metric for each agent A_i across four dimensions, expressed in a unified form as:

$$d_{\text{type},i}(x_i) = \max(0, g_{\text{type},i}(x_i)) \quad (19)$$

where $\text{type} \in \{\text{role}, \text{style}, \text{comm}, \text{coop}\}$, measuring the constraint deviation of the agent along the four dimensions of role consistency, style consistency, communication coordination, and cooperation consistency, respectively. When $d_{\text{type},i} = 0$, the constraint along that dimension is satisfied; when $d_{\text{type},i} > 0$, a deviation exists. The cooperation consistency distance is a pairwise function $d_{\text{coop},ij}(x_i, x_j)$, measuring the cooperation deviation between a pair of agents, while the remaining three types are single-agent functions.

The system-level composite distance metric measures the overall degree to which the multi-agent system deviates from the feasible region:

$$d_{\text{system}}(\mathbf{x}) = \sqrt{\sum_{i=1}^N [\alpha_i d_{\text{role},i}^2 + \beta_i d_{\text{style},i}^2 + \gamma_i d_{\text{comm},i}^2]} + \sum_{i < j} \omega_{ij} d_{\text{coop},ij}^2 \quad (20)$$

The weight parameters are dynamically adjusted based on each agent's contribution to the system.

When constraint deviation is detected, the system corrects the agent's token probability distribution through probabilistic projection. The

projection operation adopts a unified form across all constraint types:

$$P'_{type,i}(u_i) = \frac{P(u_i) \cdot \exp(-\lambda_{type,i} \cdot d_{type,i}^{pred})}{Z_{type,i}} \quad (21)$$

where $d_{type,i}^{pred}$ is the predicted constraint deviation under the given control input, $\lambda_{type,i}$ controls the correction intensity for each constraint type, and $Z_{type,i}$ is the normalization constant. This projection mechanism applies uniformly to the four constraint types—role, style, communication, and cooperation—with the correction strength along each dimension regulated by the corresponding penalty coefficient. The penalty coefficients are dynamically adjusted based on the system's operational history; the specific update rules are provided in Supplementary Note 3. Unlike the probability modification P_i in the predictive decoding of the previous section, this projection is a targeted correction for constraint deviation along specific dimensions, whereas P_i performs a global correction based on the composite measure Φ_i . The two are executed sequentially in the pipeline: predictive decoding first performs global optimization, and the monitoring module then provides targeted compensation for residual deviations.

The projections across dimensions are integrated into a unified correction signal through a composite constraint violation measure:

$$\Psi_i^{coop}(u_i) = \sum_{type} \lambda_{type,i} \cdot d_{type,i}^{pred} + \text{Payment}_i + \text{CoopCost}_i \quad (22)$$

where Payment_i is determined based on each agent's marginal contribution to constraint violations; agents with greater marginal contributions bear more correction responsibility. CoopCost_i is the cooperation cost term. When constraints of multiple agents are simultaneously violated, the system draws on the Shapley value principle from cooperative game theory for responsibility allocation and maintains cooperative incentives through transfer payments. The computation method for Shapley values and the priority rules for conflict arbitration are provided in Supplementary Note 2.

To analyze the convergence properties of the projection mechanism, the Lyapunov function of the monitoring module is defined, integrating constraint deviations across dimensions and inter-agent conflicts into a single scalar:

$$V_{coop}(\mathbf{x}) = \sum_{i=1}^N [\alpha_i d_{role,i}^2 + \beta_i d_{style,i}^2 + \gamma_i d_{comm,i}^2] + \sum_{i < j} \omega_{ij} d_{coop,ij}^2 \quad (23)$$

This function satisfies the same form of stability condition as in the previous section: under the projection strategy, it decreases monotonically along the system trajectory, guiding the system state to converge toward the feasible region.

Incentive mechanism and controllability

The preceding modules achieve constraint satisfaction and system stability. This section further enhances the long-term controllability of the system through incentive mechanisms and adaptive learning, providing agents with the motivation to actively cooperate with global optimization.

The cooperative controllability of the multi-agent system is defined as the reachability from the initial global state $\mathbf{x}(0)$ to a target state $\mathbf{x}(T) \in \mathcal{F}$, i.e., the existence of a cooperative control input trajectory $\mathbf{u}_{[0,T]}$ that drives the system state into the feasible region. The control input $u_i(t)$ includes external regulation signals such as prompt adjustments, coordination instructions, and knowledge injection.

To provide agents with the motivation to cooperate with global optimization, the system designs an incentive reward for each agent. The reward for agent i is based on its contribution to overall system performance and

collaboration effectiveness:

$$\text{Reward}_i^t = \phi_1^t \cdot \text{Contribution}_i^t + \phi_2^t \cdot \text{Coordination}_i^t + \phi_3^t \cdot \text{Welfare}_i^t + \phi_4^t \cdot \text{Fairness}_i^t \quad (24)$$

where Contribution_i^t is measured by the degree to which the agent's output improves the system's constraint satisfaction, Coordination_i^t is measured by the consistency between the agent's output and those of other agents, Welfare_i^t is measured by the agent's contribution to reducing the system-level composite distance d_{system} , and Fairness_i^t is measured by the degree of balance in resource allocation among agents. This reward design makes honest cooperation a dominant strategy for agents; the formal definition of the incentive compatibility condition is provided in Supplementary Note 2.

The system simultaneously introduces a reputation mechanism that adjusts each agent's weight and decision influence based on historical cooperation performance. Reputation values are updated using exponential moving averages: agents with consistently good performance receive higher weights, while others receive lower weights. The specific update rules and weight mappings of the reputation system are provided in Supplementary Note 3.

At the system level, this paper defines a social welfare function as the overall optimization objective:

$$\text{SocialWelfare}(\mathbf{x}, \mathbf{u}) = \sum_{i=1}^N w_i \cdot \text{Utility}_i(\mathbf{x}, \mathbf{u}) + \text{Synergy}(\mathbf{x}, \mathbf{u}) - \text{Cost}(\mathbf{u}) \quad (25)$$

where w_i is the social weight of agent i , Synergy represents the additional benefit from collaboration effects, and Cost is the control cost. This function integrates individual utility, collaboration gains, and control costs into a single optimization objective, enabling the system to balance individual performance with overall coordination.

The system achieves controllability optimization through distributed closed-loop control. When constraint deviations and coordination errors exceed thresholds, the system initiates feedback adjustment:

$$u_{coop,i}^{t+\Delta t} = u_{coop,i}^t - \eta_{coop,i}^t \cdot e_{coop,i}^t - \text{IncentiveAdjustment}_i^t \quad (26)$$

where $e_{coop,i}^t$ is the composite error signal of agent i , defined as a weighted combination of constraint distances across dimensions $e_{coop,i}^t = \sum_{type} \alpha_{type} \cdot d_{type,i}(x_i^t)$, with $d_{type,i}$ being the constraint distance metric defined in the previous section. $\eta_{coop,i}^t$ is the learning rate, dynamically adjusted based on reputation and contribution. $\text{IncentiveAdjustment}_i^t$ is the incentive correction term that guides agent behavior toward alignment with the global objective.

Integrating the incentive mechanism, reputation system, and cooperative coalition structure, the overall optimization objective for control inputs is:

$$\mathbf{u}^* = \arg \max_{\mathbf{u} \in \mathcal{U}_{coop,t}} \left[\text{SocialWelfare}(\mathbf{x}, \mathbf{u}) - \sum_{i=1}^N \text{Cost}_i(u_i) + \sum_k \text{CoalitionBenefit}_k(\mathbf{u}) \right] \quad (27)$$

where $\mathcal{U}_{coop,t}$ is the control feasible set satisfying incentive compatibility constraints, and $\text{CoalitionBenefit}_k$ is the benefit of the k -th cooperative coalition. The system allows agents to dynamically form coalitions based on task requirements and capability complementarity, with strategy sharing and benefit allocation within coalitions following the Shapley value principle. The coalition formation rules, online learning algorithms, and specific implementation of distributed coordination are provided in Supplementary Note 3.

Algorithm integration

This section summarizes the collaborative relationships and information flow among the four modules, and describes the closed-loop control process of the overall framework.

The overall framework follows a closed-loop process of “modeling–prediction–monitoring–enhancement.” The four modules each assume distinct responsibilities and interact through state information, strategy parameters, and control signals, forming a complete feedback loop.

The *constraint feasible region modeling module* provides the remaining three modules with a unified constraint representation and game structure. The joint objective function and cooperative feasible region defined by this module are directly used by the predictive decoding module. The constructed Lyapunov function $V(x(t))$ serves as a state assessment tool for the monitoring module and guides the parameter adjustments in the enhancement module.

The *collaborative predictive decoding module* solves for approximate Pareto-improving directions within the prediction horizon and generates control inputs for each agent. The output of this module provides the monitoring module with an expected behavior baseline. The model predictive control objective function simultaneously incorporates generation quality, constraint satisfaction, and a Lyapunov stability term, enabling the decoding process to balance performance and stability.

The *constraint monitoring and projection module* evaluates in real time whether the system state lies within the feasible region and corrects deviations through probabilistic projection when they are detected. This module draws on the Shapley value principle to allocate correction responsibility. Constraint violation information detected by the monitoring module simultaneously triggers learning updates in the enhancement module.

The *incentive enhancement module* continuously optimizes system performance through a reputation system and adaptive learning. The reputation values and incentive parameters of this module influence the weight settings of the utility functions in the modeling module, forming a dynamic feedback loop. The social welfare optimization objective of the enhancement module is coordinated with the Pareto-improving solution of the predictive module.

The information flow among the four modules forms a closed loop. The modeling module outputs the constraint structure and Lyapunov function; the predictive module generates control strategies accordingly; the monitoring module evaluates execution outcomes and corrects deviations; the enhancement module updates weights and incentive parameters based on long-term performance; and the updated parameters are fed back to the modeling module, initiating the next cycle. When any module detects an abnormal state, all modules respond collaboratively, restoring system operation through strategy correction and parameter adjustment.

Data availability

All data used in this study are publicly available. The datasets were sourced from the following repositories: nuScenes dataset: Available <https://www.nuscenes.org/>; GPT-Driver dataset: Available <https://www.nuscenes.org/>.

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Author contributions

Zeling Xu: Conceptualization, Methodology, Software, Experiments, Writing — Original Draft. Qianqi Zhang: Experiments, Data Analysis, Writing — Review & Editing. Dagang Li: Supervision, Theoretical Guidance, Writing — Review & Editing. Yuntao Zou: Supervision, Project Administration, Writing — Review & Editing.

Competing interests

The authors declare no competing interests.

Additional information

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